Assignment

HND IT 2nd Year, 2nd Semester 2026
Enterprise Architecture

Question: Explain the importance of using the following frameworks in systems development.

1) MVC (Model-View-Controller)

The MVC pattern separates an application into three interconnected components: the Model, which manages data and business logic; the View, which handles the presentation and user interface; and the Controller, which processes user input and coordinates between the Model and the View.

- It improves maintainability, as the user interface can be modified without affecting the underlying business logic.
- It allows parallel development, enabling front-end and back-end teams to work independently.
- It improves testability, since business logic is not mixed with presentation code.
- It promotes reusability of components across different parts of an application.

2) MVC2

MVC2 is the Java/JSP-specific implementation of the MVC pattern, developed as an improvement over the earlier Model 1 architecture, where JSP pages handled both presentation and control logic.

- It introduces a central controller (servlet) that manages all incoming requests, rather than letting individual JSP pages handle control logic.
- It solves the maintainability problems of Model 1 architecture by separating business logic from presentation.
- It became the foundation for later Java web frameworks such as Struts and Spring MVC.
- It improves the scalability and structure of large Java web applications.

3) Spring Framework

Spring is a comprehensive framework for building enterprise Java applications, providing infrastructure support so developers can focus on business logic.

- Dependency Injection (DI) reduces tight coupling between components, making applications more modular and easier to test.
- Aspect-Oriented Programming (AOP) support allows cross-cutting concerns such as logging, security, and transaction management to be handled without cluttering business logic.

• It integrates seamlessly with other technologies such as Hibernate, various databases, and messaging systems.

• It reduces boilerplate code compared to traditional Java EE development.

• Its large ecosystem (Spring Boot, Spring Security, Spring Data) speeds up development and improves productivity.

4) Hibernate

Hibernate is an Object-Relational Mapping (ORM) framework that maps Java objects directly to relational database tables.

• It eliminates the need to write repetitive SQL and JDBC code manually.

• It allows developers to work with objects instead of raw SQL queries, simplifying data access.

• It provides database portability, so switching between databases such as MySQL and Oracle requires minimal code changes.

• It offers built-in caching, lazy loading, and transaction management features.

• It reduces bugs caused by manual SQL handling and improves overall developer productivity.

5) Struts2

Struts2 is a Java web application framework built on MVC principles, designed to simplify the development of large-scale web applications.

• It provides a structured way to build web applications following the MVC2 architecture.

• It offers built-in support for form validation, internationalization, and exception handling.

• It uses interceptors to handle cross-cutting concerns, similar to the concept of AOP in Spring.

• Its tag libraries simplify JSP development and reduce repetitive coding.

• It encourages a clean separation of concerns, which is important in large, maintainable web applications.

Conclusion: Frameworks such as MVC, MVC2, Spring, Hibernate, and Struts2 play a vital role in modern systems development. They promote separation of concerns, reduce development time, improve code maintainability and testability, and provide reusable, well-tested solutions to common software development problems. Using these frameworks allows development teams to build robust, scalable, and maintainable enterprise applications more efficiently.